

Examiner reports

Q3.

While this question was generally done well there were more errors than in the first two questions. In part (a) the most common error was to include a force directed down the slope. Part (b) was done well, but a few students gave the answer 29.4 and a few used $\sin 40^\circ$ instead of $\cos 40^\circ$. Part (c) was done very well and follow through marks were awarded to those with errors arising from part (b). Part (d) was a little more demanding, with the most common error being to work with only one force. There were many good answers to part (e), but some students gave answers that were not about the forces. For example, a statement like "The box is a particle." did not relate to the forces acting on the box as was requested in the question.

Q4.

The students approached this question either using the cosine and sine rules or by finding components. The big issue for those who used the cosine rule was that some students used 30° and did not realise that in the triangle they should be considering that it was a 150° angle that was required. This approach made it very difficult for them to gain marks on the question, whereas those who had the correct angle often went on to gain full marks.

Those who used an approach based on components could do well, but there were a couple of common mistakes, typically due to getting mixed up with the values that they had obtained. A common error of this type was to use 70 instead of 104.6 in part (b) when finding the angle. All the students who had difficulties with this question would probably have been helped by better diagrams.

Q5.

Part (a) was often done very well by students, with a reasonable number of students gaining all 11 marks. There were a few issues with the force diagram, with missing or additional forces. Many students seemed to have been helped by the answer that was given for the normal reaction and this in turn enabled many to find the friction correctly. Finding the acceleration was also done well, although there were a few sign errors and some students simply used 150 instead of $150 \cos 20^\circ$. Part (b) was much more demanding and there were far fewer correct responses. The most common error was not to take account of the changed angle when finding the normal reaction and to simply continue to use the value of 243 from part (a).

In part (c) there were some good responses, but a fair number of students incorrectly stated that the tension would be greater for a greater constant speed.

Q6.

There were very many good attempts at both parts of part (a) with a lot of correct answers. However, there was quite a bit of confusion with part (b). A very common wrong answer was 686 N, and it appears that there was some confusion between the words 'resultant' and 'reaction'. There were some good solutions to part (c), although this part of the question was more demanding. A considerable range of incorrect responses were seen, including the omission of g , as in $70 \sin \alpha = 58.8$, the use of $\cos$ instead of $\sin$ or the use of the normal reaction from part (b), as in $686 = 70g \sin \alpha$.

In the second part of part (c), it was not unusual to see the 30 N force with the wrong sign

which produced the following incorrect equation: $70 g \sin \alpha = 58.8 - 30$. In part (d), there were many good answers where the candidates linked the resistance to the speed of the cyclist. However, some simply said things such as “the resistance will not be constant” and did not give a reason to support this answer.

Q7.

The candidates made very little progress with this question and did not seem to be able to deal with the resolving required.

Q8.

Most candidates answered part (a) correctly but some just wrote down the printed result.

In part (a) some candidates started with $m \frac{dv}{dt} = -9.8v$ but found that this did not give the printed result, and just invented a term in g . This was not accepted. Some were penalised for ignoring the m term or for changing signs.

In part (b) candidates were expected to use $\int \frac{1}{v-5} dv = -1.96t$; some split $\int \frac{1}{v-5} dv$ into $\int \frac{1}{v} - \frac{1}{5} dv$. However there were many good answers to this part with both the integrations required being successfully completed. The difficulty which some candidates had was in the simplification needed to convert $\ln(v-5) = -1.96t + \ln 2$ into $v-5 = 2e^{-1.96t}$ which often became $v-5 = e^{-1.96t} + 2$

Q9.

While there were a number of good solutions to this question, there were also a very significant number of weak attempts. In part (a), many candidates could write down either a correct horizontal or vertical equation, but then were unable to derive the required answer correctly. Some could get one equation but not the other. In some cases the equations were confused, for example containing both a and g in the same equation. In part (b), there were again a number of correct solutions. The most common error here was to work on the assumption that P would also be zero. The candidates who answered correctly produced a variety of solutions, which included finding an expression for Q and starting again as if there was only one string.

Q10.

Most candidates answered parts (a), (b) and (c) well, although some did get an answer of $-4.5j$ for part (b).

In part (c), the candidates who clearly stated and used a constant acceleration equation usually made a good start and completed the question, although there were a few arithmetic errors.

In part (e) many candidates did not attempt to find the magnitude of the force.

Part (f) was generally found more difficult with only a few good attempts. Several candidates did not attempt this part of the question.

Q11.

Parts (a) and (b) were generally done well by the candidates. A few candidates subtracted the vectors in part (a) instead of adding them. Also some tried to find the magnitude of each vector in part (a) and then lost several marks because they had very confused working.

In part (c) there were many attempts which involved the division of vectors. For example

working such as: $m = \frac{9\mathbf{i} + 12\mathbf{j}}{1.5\mathbf{i} + 2\mathbf{j}} = 6\mathbf{i} + 6\mathbf{j}$. Candidates should be advised to consider the components involved and not to attempt to divide vectors.

There were a number of errors that appeared in part (d). One was to include an initial velocity, which was equal to the resultant force on the particle. A few candidates used incorrect formulae. In the final part of the question a large number of candidates gave the vector $3\mathbf{i} + 4\mathbf{j}$ as their final answer and did not find the magnitude in order to get the distance as requested.

Q12.

The force diagram in part (a) was often drawn well, but the usual errors of extra forces and missing labels or arrows were seen. Part (b) was done very well, with the printed answer helping almost all of the candidates to gain full marks. The responses to part (c) were very varied, with some solutions excellent and some very poor. One fairly common error was to form an equation with a sign error, like $F - 5g \sin 40^\circ = 5 \times 0.8$. Other more serious errors included using the normal reaction force instead of the weight component parallel to the slope, and simply assuming that the friction was equal to 4. Part (d) of the question was found to be very difficult, with very few candidates giving acceptable answers. The majority of candidates stated that the coefficient of friction would increase.

Q13.

In part (a) there were many good diagrams, but also a number that included extra forces, especially parallel to the slope. On some diagrams the arrows or labels were missing. Part (b) was done well and the vast majority of candidates did show enough working to gain all of the marks. Some candidates clearly used the given answer to correct their working. Part (c) was more demanding. When finding the acceleration, the main errors were not dividing the resultant force by the mass; making a sign error to obtain $a = -0.171$; not considering both forces, to produce equations like $100a = 68.4$; and not resolving the weight correctly.

In the last part, there were some good explanations, but there were also a lot that were not acceptable. Some candidates wrote about the resistance being related to the acceleration rather than the speed or velocity. A few also thought that the resistance would depend on the slope.

Q14.

Parts (a)(i), (ii), (v) and (b) proved routinely successful for well prepared and accurate candidates. Parts (a)(iii) and (iv) tested the understanding of the forces in the static situations and proved highly discriminating.

Q15.

Again, this was answered very well, often scoring full marks.

Q16.

Diagrams in part (a) were clear and mostly correct, with occasional marks lost due to missing arrows or a missing or incorrect force, with the weight being given as 'g' as the most frequent. Parts (b) and parts (c) were also good, but few scored well in part (d), with one or even two forces missing from the equation of motion, the weight component being the most frequent omission.

Q17.

There was a good number of correct force diagrams. Some candidates added one or both of the components of the tension to their force diagram. In some cases these appeared as if they were additional forces. Parts (b) and (c) were done well by most candidates who seemed to have benefitted from the printed answers. There were some candidates who did not gain full marks, because they did not show sufficient working. In part (d), many candidates obtained the correct value. Some candidates used the two printed answers as expected, if they had not been able to obtain the two values for themselves. There were some very good answers to part (e), but there were also quite a few confused ones as well, with many candidates finding it hard to give a good answer.

Q18.

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Q19.

As a whole candidates seemed to find this question more demanding. While some made good attempts there were a number of candidates that showed weaknesses in resolving or that did not recognise that resolving was required. In part (b), there was a number of candidates who omitted the weight from their solutions, simply summing the vertical components of the two tensions.

Q20.

A variety of force diagrams were seen on the candidates' scripts. Some were very good, but errors included the omission of forces and showing the friction force acting up the slope. In part (b) the candidates were clearly helped by the printed answer and there were many correct responses. In some cases the candidates did not show enough working to gain full marks. Part (c) proved to be very challenging, with only a very small number of candidates getting the correct coefficient of friction. The main problem was that the candidates did not include all of the terms that were needed in their equations of motion.

Q21.

Diagrams were mostly very good with only occasional lapses in the direction of the normal reaction force or in poor labelling of forces. Part (a) proved popular and successful for most candidates. In part (b)(i) a number failed to appreciate that the forces involved included the weight component, but many went on to be successful in part (ii).

Q22.

Many candidates were fully successful in part (a). In part (b), however, complete equations of motion were rare and components were often inappropriately applied: either incorrect or used with inappropriate forces.

Q23.

Parts (a)(i) and (ii) were often well done, although there was some use of the 'straight string' method, and some lack of understanding of relevant forces acting on the particle on the horizontal surface.

Part (a)(iii) proved discriminating as expected, but part (b) revealed a lack of understanding of the two stages of the motion.

Q24.

The significance of the direction of the resultant force was lost on many. Again answers were often disappointing, with errors finding components and confusion with directions of forces.

Q25.

A surprising number found the static part of this question very demanding, with relatively few correct responses to parts (a)(i), (iii) and (iv). Part (b) proved more successful.

Q26.

Many candidates attempted this question satisfactorily, realising that the sum of the components of the three given forces was zero parallel to Ox , but non-zero parallel to Oy . Resulting equations were often fully correct and accurate arithmetical work followed. Some errors in selecting the correct component or sign were seen, but the most common mistake was failure to find the magnitude of the resultant and thereby complete part (b).

Q27.

This question was answered fairly well by the majority of candidates. On the whole the force diagrams were quite good, but a number of classic errors were observed. For example including a force in the direction of motion and putting the normal reaction vertical. The rest of part (a) was answered very well, with the candidates appearing to have been helped by the printed answers.

The answers to part (b) were much more variable. There were quite a few very good answers, which included clear concise explanations.

Q28.

The candidates were more successful with this question than had been expected.

Part (a) was quite well answered. A reasonable number of candidates formed the two equations of motion correctly and then went on to obtain the required value. Some candidates did not resolve the weight of the particle on the slope. A small number of candidates did not form an equation of motion for each particle.

Those candidates who realised that the acceleration was zero did well in part (b). There were quite a few candidates who tried to use the acceleration from part (a) when tackling part (b).

Q29.

Parts (a) and (b) were generally done well. In part (b), the printed answer helped many candidates, but some could not put together a convincing argument. Part (c) was more challenging. Some candidates did resolve horizontally and apply the friction law, but then got into difficulties solving the resulting equation. Other candidates did not take both of the steps needed to produce an equation for T and made very little progress. There was also some evidence of candidates using $\sin 30^\circ$ instead of $\cos 30^\circ$ when resolving horizontally.

Q30.

The force diagram asked for in this question was drawn fairly well, but the usual errors of forces in the direction of motion and missing forces were present on some scripts. Parts (b) and (c) were done fairly well and it appeared that many 'candidates' had been helped by the printed answer in part (b). There were some candidates who did not realise that they should consider the tension in the rope when finding the normal reaction. Part (d) was found to be very much more demanding, with relatively few correct responses. Candidate typically omitted one term from their equation or failed to solve correctly for T .

Q31.

This question was done quite well on the whole. The major source of errors was resolving incorrectly. The fact that the friction force was given did help some candidates to work towards the correct answer.

Q32.

The candidates found this question more difficult. Some treated it like a connected particles question and introduced an acceleration even though the system was in equilibrium. Candidates who persisted in using this approach made very little progress. While a number of candidates did produce complete solutions, there were a variety of errors, often involving resolving, that were made by a number of candidates.

Q33.

Part (a) was generally done well. Some candidates made arithmetic errors and a small number misquoted the constant acceleration equation. The force diagrams were drawn very well. Most of the candidates found it easy to calculate the magnitude of the friction force, although a small number took the reaction force as being equal to the weight. Many candidates had difficulty finding the tension in the rope.

Q34.

There were a number of good responses to this question. Some parts caused the

candidates a little more difficulty than the earlier questions. In part (a), some candidates included additional forces. Parts (b) and (c) were generally well done, the only significant problem was that some candidates used the wrong angle when finding the components of the weight and so ended up with their answers to parts (b) and (c) interchanged. This caused subsequent problems in part (d). However, most candidates made a good attempt at part (d). Part (e) was more challenging. Many candidates felt that the 14 kg would just slide down the slope. Some argued that the friction and weight would be proportional to the mass and so it would stay still. Quite a number repeated all the earlier calculations to obtain the same coefficient of friction and often concluded that the block would remain at rest.



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